

MIP_TESTING_STRATEGY.md

Mainframe Intelligence Platform

Testing Strategy

Version: 1.0

Purpose

Ensure reliability and correctness.

Testing is mandatory.

Testing Pyramid

Unit Tests ↑

Integration Tests ↑

End-to-End Tests

Unit Tests

Required for:

- Parsers
- Models
- Graph Components
- Services

Coverage Goal:

90%

Integration Tests

Validate:

Discovery → Metadata

Metadata → Graph

Graph → Queries

End-to-End Tests

Validate:

Repository Scan

↓

Metadata Extraction

↓

Knowledge Graph

↓

Insights

Test Categories

Parser Tests

Validate:

- PROGRAM-ID
 - CALL
 - COPY
 - EXEC SQL
-

Metadata Tests

Validate:

- Entity creation
 - Validation rules
 - Schema evolution
-

Graph Tests

Validate:

- Node creation
 - Edge creation
 - Traversal
-

Performance Tests

Validate:

- Large repository scans
 - Graph construction
 - Query performance
-

Test Data

Store under:

tests/fixtures/

Use:

- Sample COBOL
 - Sample JCL
 - Sample Copybooks
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CI Requirements

Every PR:

- Lint
- Unit Tests
- Integration Tests

Must pass before merge.

Success Criteria

80% minimum coverage.

90% target coverage.

No critical defects.